COMP9418 – Advanced Topics in Statistical Machine Learning

W7 – Continuous Latent Variables

Instructor: Amir Dezfouli

School of Computer Science and Engineering

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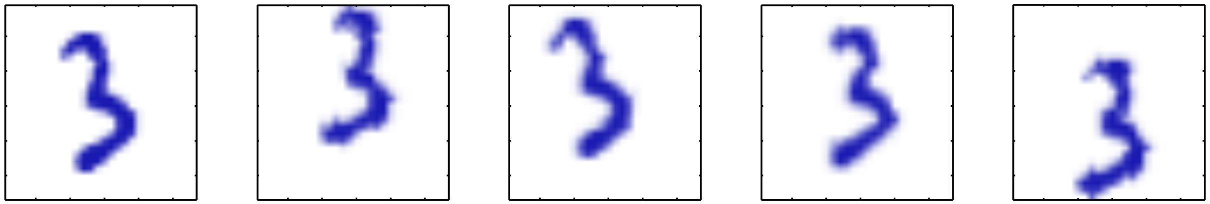
Acknowledgements

Material derived from:

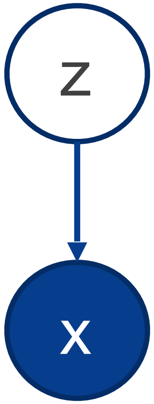
- [Bishop, PRML, 2006] Pattern Recognition and Machine Learning, Christopher Bishop, 2006
  - Some figures taken from this book
- [Murphy, MLaPP, 2012] Machine Learning: A Probabilistic Perspective, Kevin P. Murphy, 2012
  - Some figures taken from this book
- Other resources (as mentioned in the slides)

Why Continuous Latent Variables

- Data may lie close to a manifold of much lower dimensionality



- Same digit randomly displaced and rotated within larger image
- 10,000-dimensional → intrinsic dimensionality of 3
  - Rotation and translation parameters are *latent variables*



- Mixture models use a single latent variable to generate the observations
  - K latent variables using one-hot-encoding
  - These are mutually exclusive
  - Representational power is still limited

Dimensionality Reduction

- Learning in high-dimensional spaces may be very hard
- We can use low-dimensional representations for visualisation
- These representations can be more discriminative

Aims (1)

This lecture will allow you to understand some popular probabilistic models with continuous latent variables and how to carry out inference in these models. Following it you should be able to:

- Carry out dimensionality reduction with principal component analysis (PCA)
- View PCA from a probabilistic perspective yielding probabilistic principal component analysis (PPCA), and estimate its parameter via direct likelihood maximisation and the EM algorithm
- View Factor Analysis as a generalisation of PPCA and be able to estimate its parameters via EM
- Understand the fundamental differences between PPCA and FA

Aims (2)

- Fit an independent component analysis (ICA) model for blind source separation of signals and see this model from a probabilistic modelling perspective
- Understand the differences between ICA, FA and PPCA and apply them to suitable machine-learning problems requiring continuous latent variables

Outline

I. Principal Component Analysis (PCA) Revisited

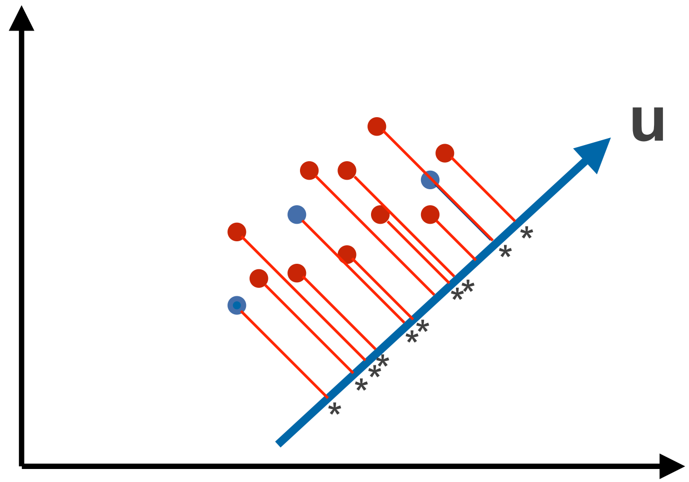
II. Probabilistic Principal Component Analysis (PPCA)

III. Factor Analysis (FA)

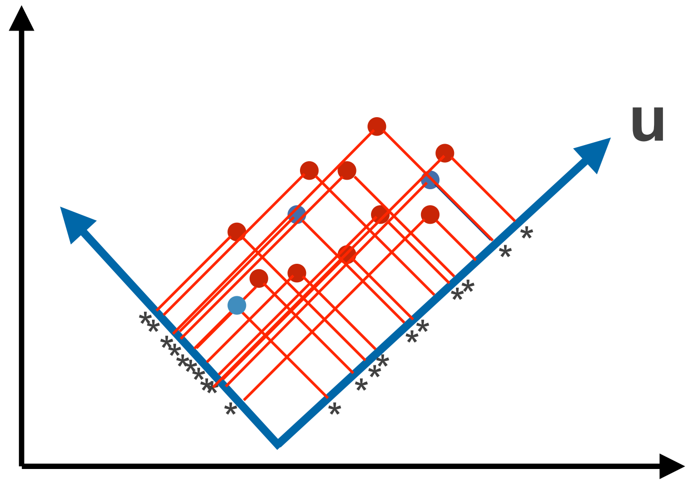
IV. Independent Component Analysis (ICA)

I. Principal Component Analysis (PCA) Revisited

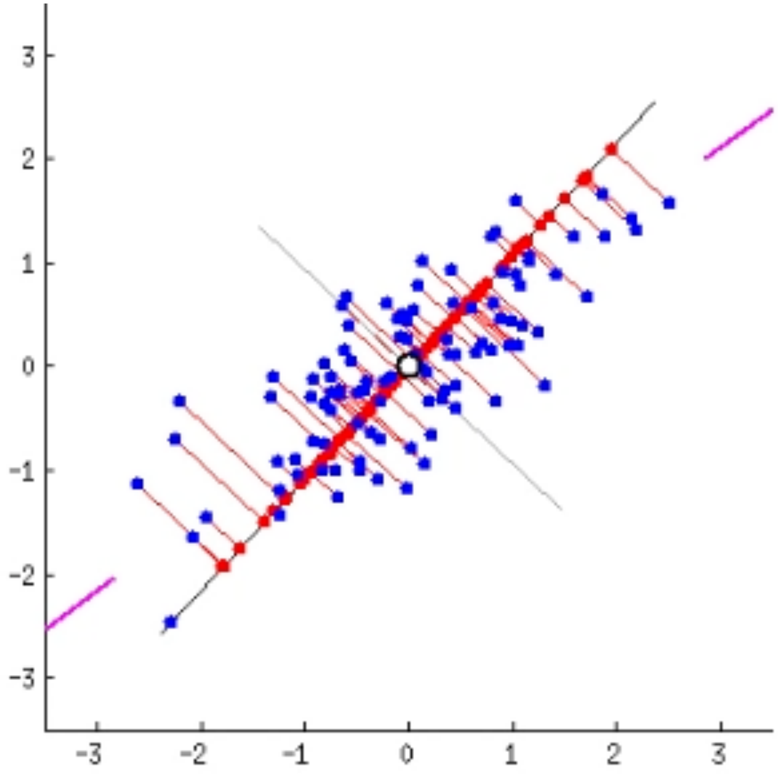
Principal Component Analysis (PCA)



Principal Component Analysis (PCA)

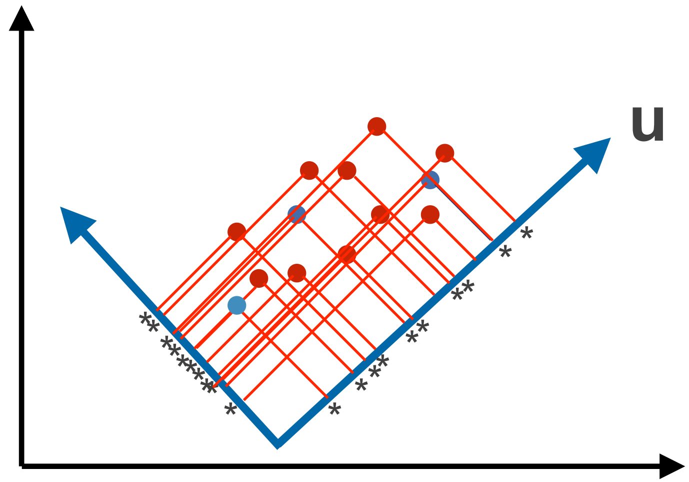


Principal Component Analysis (PCA)



from: https://gist.github.com/anonymous/7d888663c6ec679ea65428715b99bfdd

Principal Component Analysis (PCA)

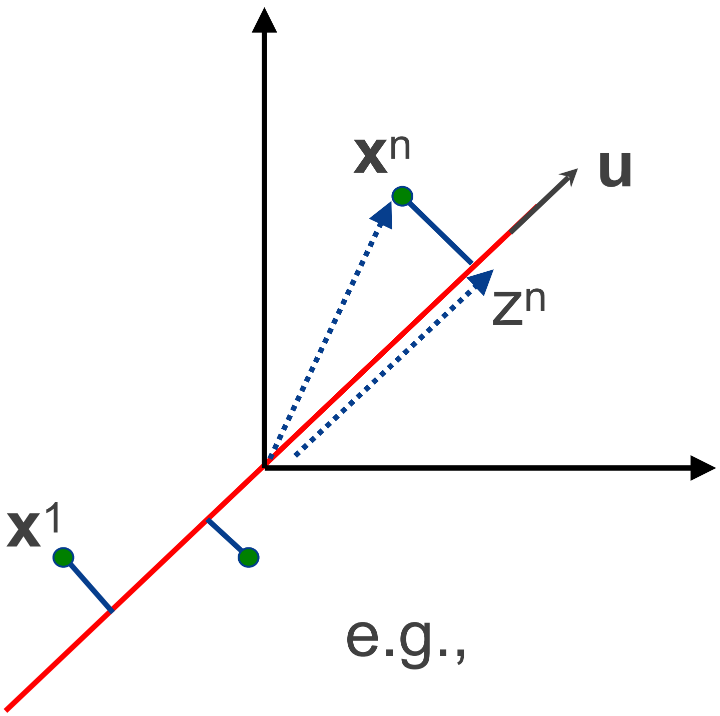


u should be in the direction that the projected datapoints have **maximum variance**

Principal Component Analysis (PCA)

Consider a projection into a one-dimensional space spanned by the unit vector **u** with $\mathbf{u}^T\mathbf{u} = 1$

$$z^n = \mathbf{u}^T \mathbf{x}^n$$



Each datapoint is projected into a scalar value

The mean of the projected data is:

$$\bar{z} = \mathbf{u}^T \bar{\mathbf{x}}$$

Data mean

And the variance:

$$\sigma^2(\mathbf{u}) = \frac{1}{N}\sum_{n=1}^{N}(z^n - \bar{z})^2$$

$$= \frac{1}{N}\sum_{n=1}^{N}\left\{\mathbf{u}^T\mathbf{x}^n - \mathbf{u}^T\bar{\mathbf{x}}\right\}^2$$

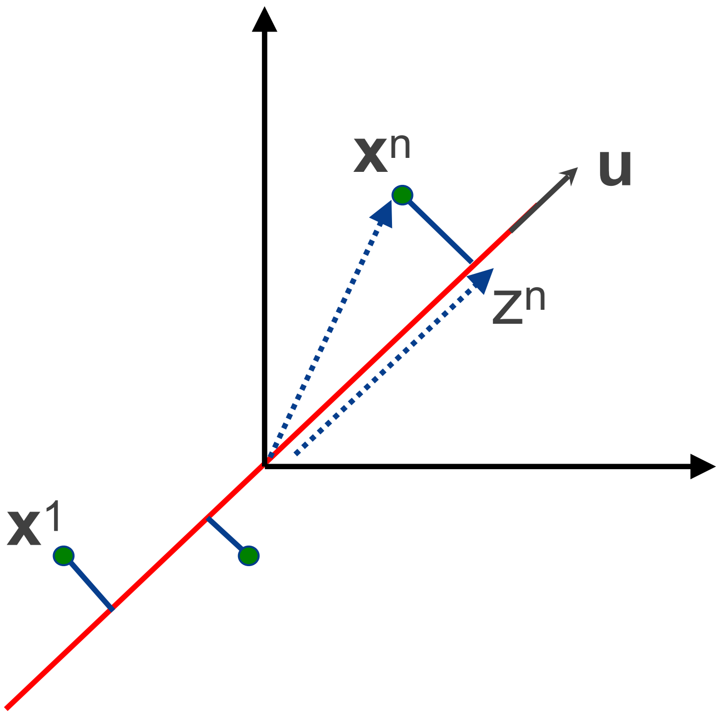
Data covariance

$$= \mathbf{u}^T \mathbf{S} \mathbf{u}$$

e.g.,

$$\mathbf{S} = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

Principal Component Analysis (PCA)



$$\sigma^2(\mathbf{u}) = \mathbf{u}^{\mathbf{T}}\mathbf{S}\mathbf{u}$$

We want to maximise $\sigma^2(\mathbf{u})$ wrt **u** subject to $\mathbf{u}^T\mathbf{u} = 1$

Why this constraint?

Using Lagrange multipliers:

$$\mathbf{u}^{\mathbf{T}}\mathbf{S}\mathbf{u} - \lambda(1 - \mathbf{u}^T\mathbf{u})$$

Derivative w.r.t to **u:**

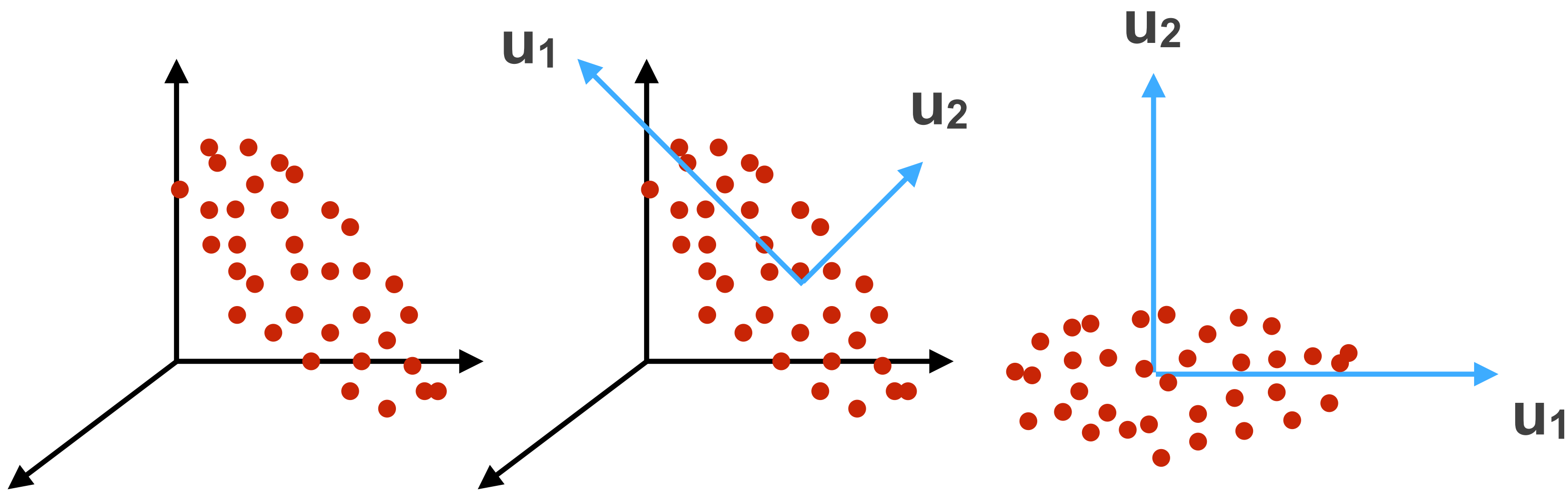
u is an eigenvector

$$\mathbf{S}\mathbf{u} = \lambda\mathbf{u}$$

$$\mathbf{u}^T\mathbf{S}\mathbf{u} = \mathbf{u}^T\lambda\mathbf{u} = \lambda$$

- This means that **u** is the eigenvector corresponding to the largest eigenvalue (λ) of **S**. **u** is known as the first principal component
- λ is the amount of variance explained by **u**

Principal Component Analysis (PCA)



- We can incrementally add more principal components so that the projected variance is maximised

- The PCA solution to our original problem is given by the K eigenvectors $\mathbf{u}^1,\ldots,\mathbf{u}^K$ corresponding the the K largest eigenvalues of $\mathbf{S}$, $\lambda_1,\ldots,\lambda_K$.

Principal Component Analysis (PCA)

Problem Definition

- Consider a set of N datapoints $\{\mathbf{x}^{(n)}\}_{n=1}^{N}$ with $\mathbf{x} \in \mathbb{R}^D$.
- Our goal is to find a low-dimensional linear representation $\mathbf{z}^{(n)} \in \mathbb{R}^K$ with K < D that explains the data well.

Principal Component Analysis (PCA)

Input: D-dimensional data $\{\mathbf{x}^{(n)}\}_{n=1}^{N}$, K

1. Compute sample mean $\bar{\mathbf{x}}$ and covariance $\mathbf{S}$
2. Find K eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_K$ corresponding to the largest K eigenvalues $\lambda_1, \ldots, \lambda_K$ and construct the matrix

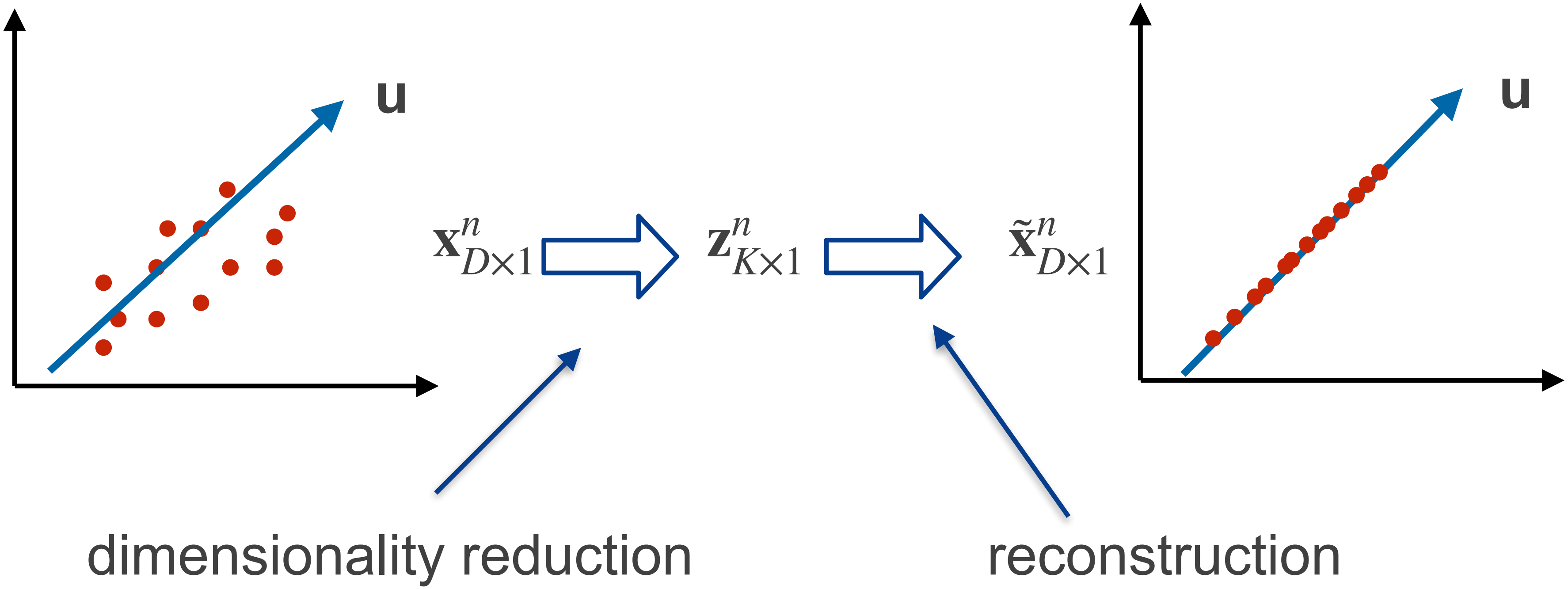
$$\mathbf{U} = (\mathbf{u}_1, \ldots, \mathbf{u}_K)_{D \times K}$$

3. Output: Lower dimensional representation is given by:

$$\mathbf{z}^n = \mathbf{U}^T(\mathbf{x}^n - \bar{\mathbf{x}})$$

$$\mathbf{z}^n = \begin{bmatrix} \mathbf{u}_1^T(\mathbf{x}^n - \bar{\mathbf{x}}) \\ \vdots \\ \mathbf{u}_K^T(\mathbf{x}^n - \bar{\mathbf{x}}) \end{bmatrix}_{K \times 1}$$

Principal Component Analysis (PCA)



minimum reconstruction error and minimum variance give use same solution

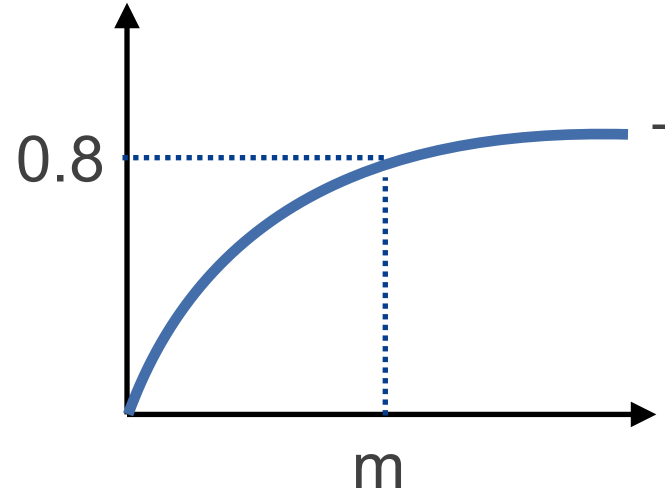
Principal Component Analysis (PCA)

How much variance have we explained by selecting K components?

$$\sum_{k=1}^{K} \lambda_k$$

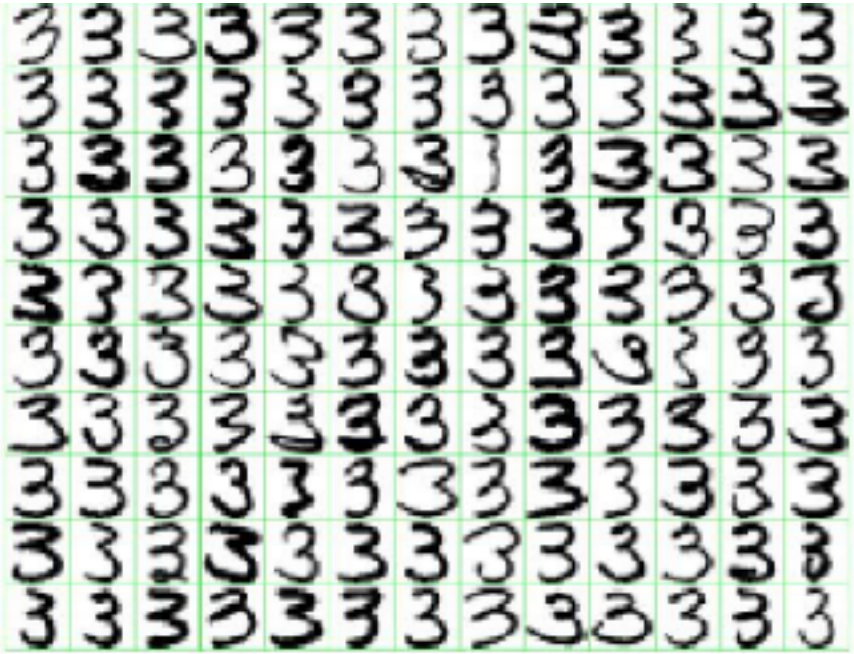
variance explained by eigenvector *k*

- We can use this to establish a criterion to select K in an unsupervised setting.

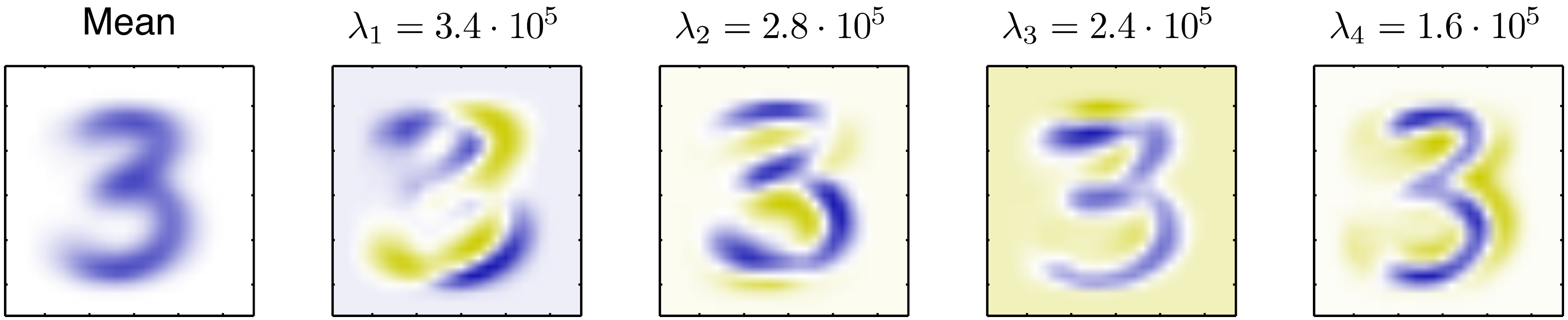


$$\frac{\sum_1^m \lambda_i}{\sum_1^D \lambda_i} \leq 1$$

PCA Example



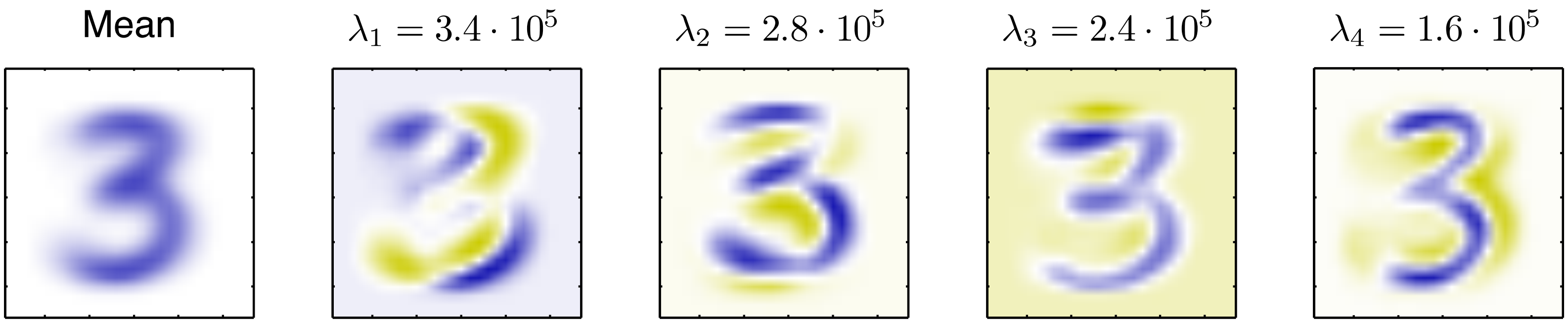
Mean and top eigenvectors



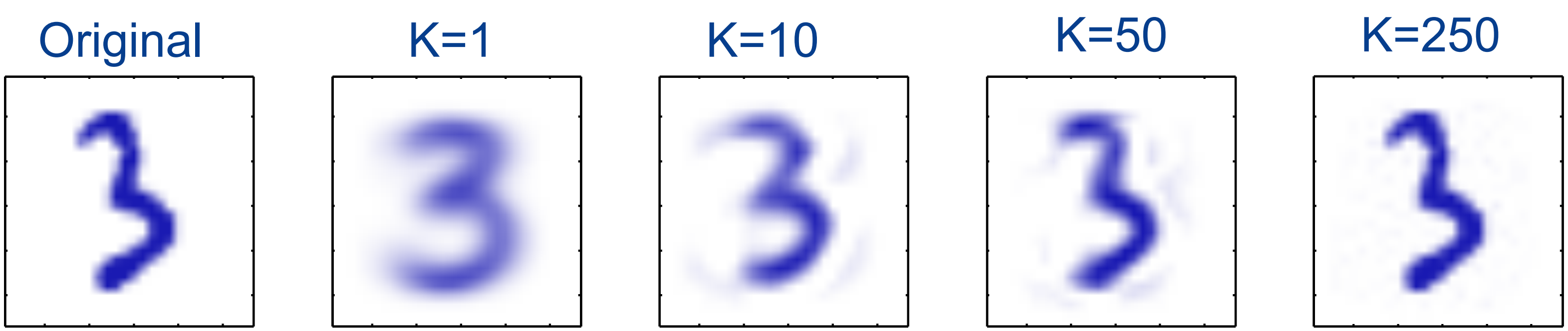
(using MNIST dataset with only digit=3)

PCA Example

Mean and top eigenvectors



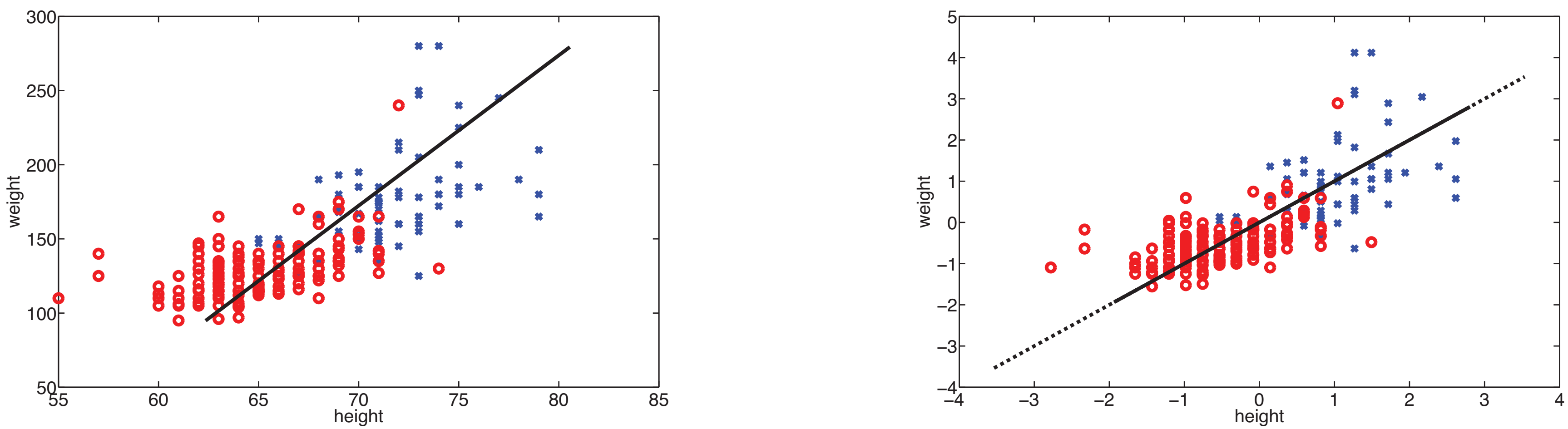
Reconstruction



(using MNIST dataset with only digit=3)

PCA and measurement scales

- PCA can be "misled" by directions in which the variance is high merely because of the measurement scale.



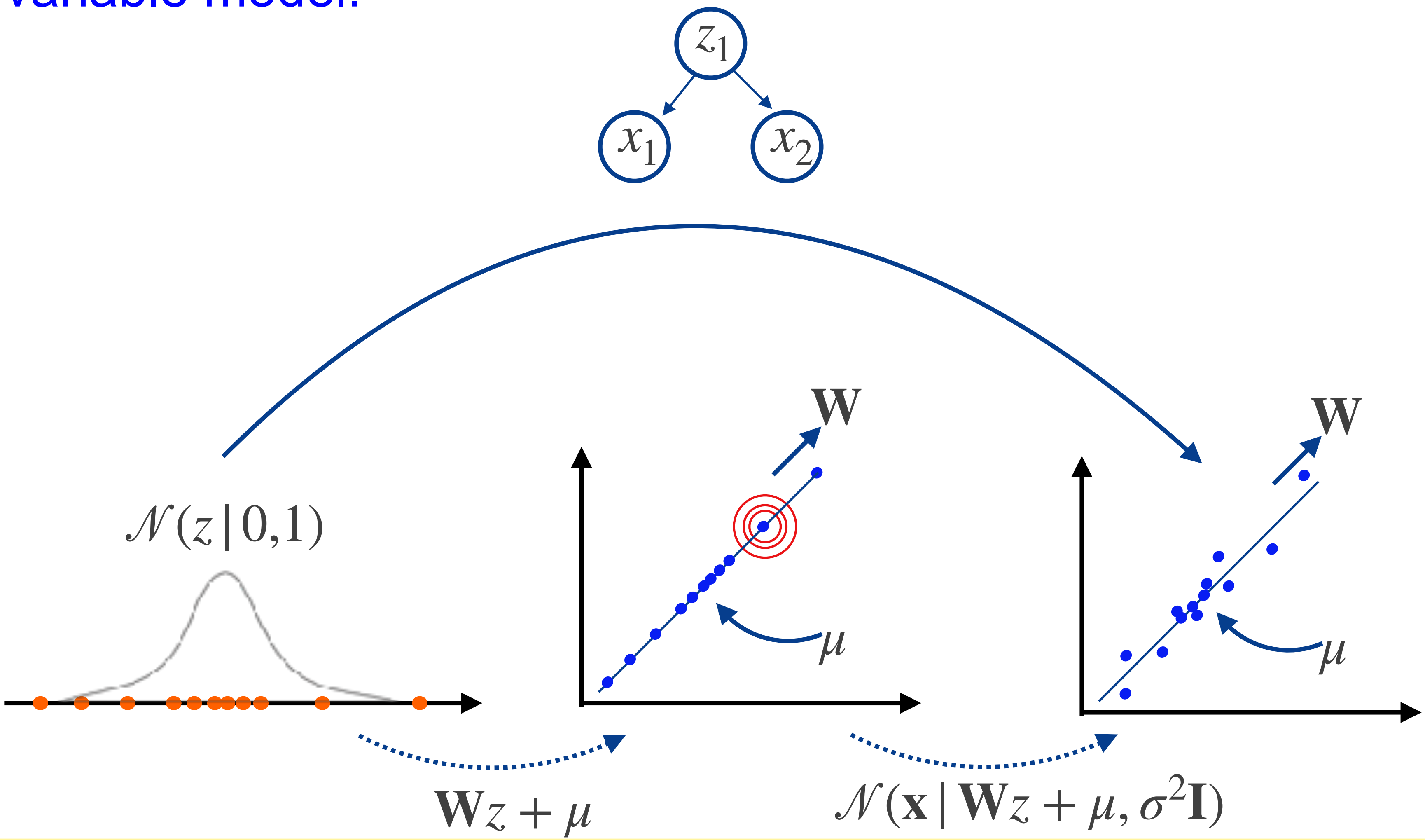
(a) (b)

- It is common to standardise data before PCA
- Or we can work the correlation matrix instead of covariance matrix

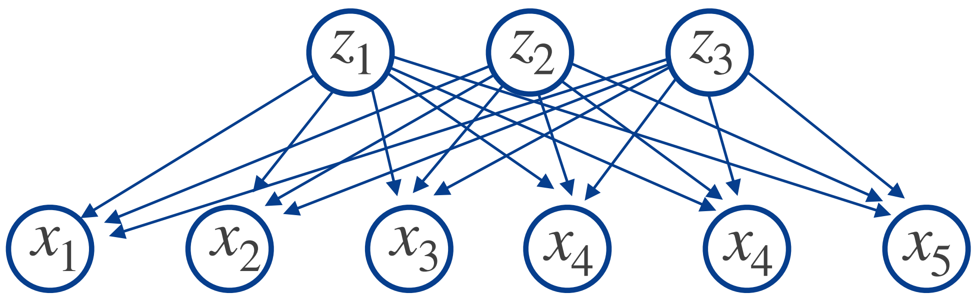
II. Probabilistic Principal Component Analysis (PPCA)

Probabilistic Principal Component Analysis (PPCA)

In fact, PCA can be derived from a proper probabilistic latent variable model:



Probabilistic Principal Component Analysis (PPCA)



We observe $\mathbf{x} \in \mathbb{R}^D$ and assume that it is generated by a latent low-dimensional vector $\mathbf{z} \in \mathbb{R}^K$:

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{z}|\mathbf{0}, \mathbf{I}), \qquad p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}|\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I})$$

- The D × K matrix **W**, the D-dimensional bias vector **μ** and the noise variance σ^2 are the model parameters

Probabilistic Principal Component Analysis (PPCA)

observations (D dimensional)

$$\mathbf{X} = \{\mathbf{x}^{(n)}\}_{n=1}^{N}$$

$$\mathbf{Z} = \{\mathbf{z}^{(n)}\}_{n=1}^{N}$$

latent variables (K dimensional)

$$\boxed{\begin{aligned} p(\mathbf{z}) &= \mathcal{N}(\mathbf{z}|\mathbf{0}, \mathbf{I}), \\ p(\mathbf{x}|\mathbf{z}) &= \mathcal{N}(\mathbf{x}|\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I}) \end{aligned}}$$

How to learn model parameters μ, **W**, σ^2?

we need to marginalise latent variables

$$p(\mathbf{x}) = \int p(\mathbf{x}\,|\,\mathbf{z}, \mathbf{W}, \mu, \sigma^2)p(\mathbf{z})d\mathbf{z}$$

analytical solution → $= \mathcal{N}(\mathbf{x}\,|\,\mu, \mathbf{W}^T\mathbf{W} + \sigma^2\mathbf{I})$

Probabilistic Principal Component Analysis (PPCA)

Maximum Likelihood estimates

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}),$$

- Assuming iid observations, the data log-likelihood is:

$$\mathscr{L}(\mathbf{W}, \mu, \sigma^2) = p(\mathbf{X} \,|\, \mathbf{W}, \mu, \sigma^2)$$

$$= \sum_{n=1}^{N} p(\mathbf{x}^n \,|\, \mathbf{W}, \mu, \sigma^2)$$

- and Maximizing $\mathcal{L}$ wrt **μ**, **W** and **σ**² we obtain:

$$\widehat{\boldsymbol{\mu}}_{\mathrm{ML}} = \bar{\mathbf{x}} \quad \widehat{\mathbf{W}}_{\mathrm{ML}} = \mathbf{U}(\Lambda - \sigma^2_{\mathrm{ML}})^{1/2}\mathbf{R} \qquad \widehat{\sigma}^2_{\mathrm{ML}} = \frac{1}{D-K}\sum_{i=K+1}^{D} \lambda_i$$

Sample mean

Ordered eigenvalues and eigenvectors of data covariance

Rotation matrix, $\mathbf{R}\mathbf{R}^T = \mathbf{I}$

Average variance of Discarded dimensions

How to Do Dimensionality Reduction

$$\boxed{\begin{aligned} p(\mathbf{z}) &= \mathcal{N}(\mathbf{z}|\mathbf{0}, \mathbf{I}), \\ p(\mathbf{x}|\mathbf{z}) &= \mathcal{N}(\mathbf{x}|\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I}) \end{aligned}}$$

$$\Downarrow$$

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}),$$

$$\Downarrow$$

$$\hat{\mathbf{W}}_{\text{ML}}, \hat{\mu}_{\text{ML}}, \hat{\sigma}^2_{\text{ML}}$$

$$\Downarrow$$

predictive distribution $p(\mathbf{z}\,|\,\mathbf{x}) = p(\mathbf{x}\,|\,\mathbf{z})p(\mathbf{z})/p(\mathbf{x})$

$$p(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}|\mathbf{M}^{-1}\mathbf{W}^T(\mathbf{x} - \boldsymbol{\mu}), \sigma^{-2}\mathbf{M}^{-1}), \quad \mathbf{M}_{K\times K} = \mathbf{W}^T\mathbf{W} + \sigma^2\mathbf{I}$$

How to Do Dimensionality Reduction

$$p(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}|\mathbf{M}^{-1}\mathbf{W}^T(\mathbf{x}-\boldsymbol{\mu}), \sigma^{-2}\mathbf{M}^{-1}), \quad \mathbf{M} = \mathbf{W}^T\mathbf{W} + \sigma^2\mathbf{I}.$$

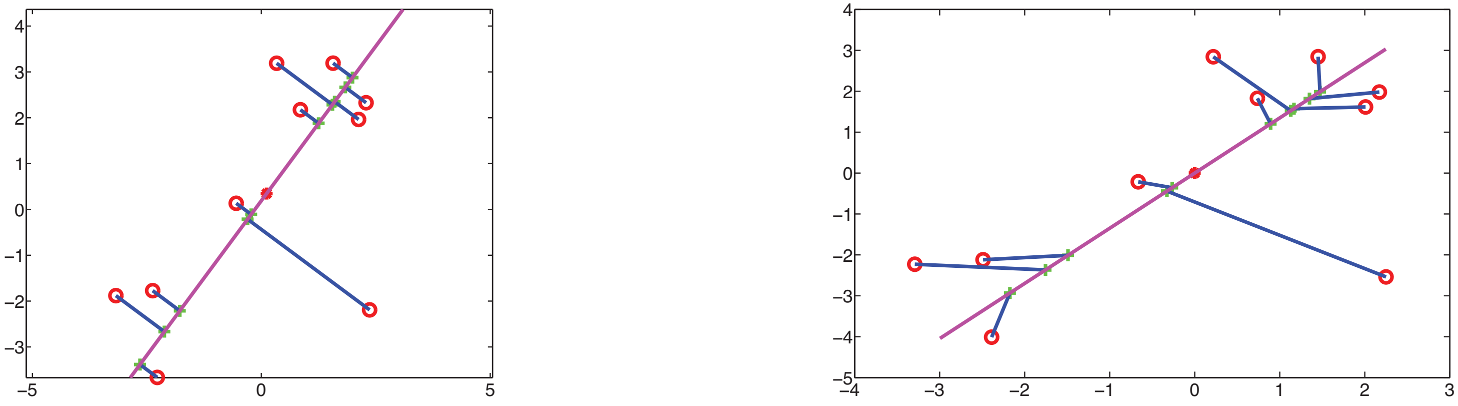
$$\mathbb{E}[\mathbf{z}|\mathbf{x}] = (\widehat{\mathbf{W}}_{\mathrm{ML}}^T\widehat{\mathbf{W}}_{\mathrm{ML}} + \widehat{\sigma}_{\mathrm{ML}}^2\mathbf{I})^{-1}\widehat{\mathbf{W}}_{\mathrm{ML}}^T(\mathbf{x}-\bar{\mathbf{x}})$$

and taking the limit $\sigma^2 \to 0$ we obtain PCA, an orthogonal projection of the datapoints into the latent space:

$$\mathbb{E}[\mathbf{z}\,|\,\mathbf{x}] \to \Lambda^{-1/2}\mathbf{U}^T(\mathbf{x}-\bar{\mathbf{x}})$$

PCA vs PPCA projections

$$\mathbb{E}[\mathbf{z}|\mathbf{x}] = (\widehat{\mathbf{W}}_{\mathrm{ML}}^T \widehat{\mathbf{W}}_{\mathrm{ML}} + \widehat{\sigma}_{\mathrm{ML}}^2 \mathbf{I})^{-1} \widehat{\mathbf{W}}_{\mathrm{ML}}^T (\mathbf{x} - \bar{\mathbf{x}})$$



(a)

(b)

$\sigma^2 \longrightarrow 0$

orthogonal projection

$\sigma^2 > 0$

projections are shrunk toward prior mean (0,0)

EM for PPCA

$$\boxed{\begin{aligned} p(\mathbf{z}) &= \mathcal{N}(\mathbf{z}|\mathbf{0}, \mathbf{I}), \\ p(\mathbf{x}|\mathbf{z}) &= \mathcal{N}(\mathbf{x}|\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I}) \end{aligned}}$$

$$\Downarrow$$

$$\hat{\mathbf{W}}_{\text{EM}}, \hat{\mu}_{\text{EM}}, \hat{\sigma}^2_{\text{EM}} \Leftarrow \text{EM} \Leftarrow p(\mathbf{x}) = \int p(\mathbf{x}\,|\,\mathbf{z}, \mathbf{W}, \mu, \sigma^2)p(\mathbf{z})d\mathbf{z}$$

$\Downarrow$ marginalising **z**

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}\,|\,\mu, \mathbf{W}^T\mathbf{W} + \sigma^2\mathbf{I})$$

$\Downarrow$ maximum likelihood

$$\hat{\mathbf{W}}_{\text{ML}}, \hat{\mu}_{\text{ML}}, \hat{\sigma}^2_{\text{ML}}$$

EM for PPCA

- PPCA is a latent variable model hence we can use EM for parameter estimation
  - Computational advantages over direct MLE for high dimensionality
  - Can be extended to factor analysis
  - Can handle missing data
- Let $\boldsymbol{\theta} = \{\boldsymbol{\mu}, \mathbf{W}, \sigma^2)$: $\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) = \mathbb{E}_{p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta}^{\text{old}})}[\log p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})]$

1. E-step: Compute $\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}})$
   - Compute posterior p(**Z** | **X**, θ^{old}) as a function of (old) parameters
   - Expected sufficient statistics
2. M-step: Optimize $\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}})$ wrt θ:

   $\boldsymbol{\theta}^{\text{new}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, \mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}})$: Parameters at new iteration

EM for PPCA: E-step

- $\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\mathrm{old}})$ can be computed using the expected sufficient statistics:

$$\mathbf{m}_n = \mathbb{E}_{p(\mathbf{z}^{(n)} | \mathbf{x}^{(n)}, \boldsymbol{\theta}^{\mathrm{old}})}\left[\mathbf{z}^{(n)}\right] = \mathbf{M}_{\mathrm{old}}^{-1}\mathbf{W}_{\mathrm{old}}^{T}(\mathbf{x}^{(n)} - \overbrace{\boxed{\bar{\mathbf{x}}}}^{\widehat{\boldsymbol{\mu}}_{\mathrm{ML}}})$$

$$\boldsymbol{\Sigma}_{\mathrm{n}} = \mathbb{E}_{p(\mathbf{z}^{(n)} | \mathbf{x}^{(n)}, \boldsymbol{\theta}^{\mathrm{old}})}\left[(\mathbf{z}^{(n)})(\mathbf{z}^{(n)})^{T}\right] = \sigma_{\mathrm{old}}^{2}\mathbf{M}_{\mathrm{old}}^{-1} + \mathbf{m}_n\mathbf{m}_n^{T}$$

where $\quad \mathbf{M}_{\mathrm{old}} = \mathbf{W}_{\mathrm{old}}^{T}\mathbf{W}_{\mathrm{old}} + \sigma_{\mathrm{old}}^{2}\mathbf{I}$

EM for PPCA: M-step

- We keep the expected sufficient statistics fixed and solve:

$$\boldsymbol{\theta}^{\text{new}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, \mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}})$$

Obtaining:

$$\mathbf{W}_{\text{new}} = \left[\sum_{n=1}^{N} (\mathbf{x}^{(n)} - \bar{\mathbf{x}})\mathbf{m}_n^T\right] \left[\sum_{n=1}^{N} \boldsymbol{\Sigma}_{\text{n}}\right]^{-1}$$

$$\sigma^2_{\text{new}} = \frac{1}{ND} \sum_{n=1}^{N} \{ \|\mathbf{x}^{(n)} - \bar{\mathbf{x}}\|^2 - 2\mathbf{m}_n^T \mathbf{W}_{\text{new}}^T (\mathbf{x}^{(n)} - \bar{\mathbf{x}}) + \operatorname{tr}\left(\boldsymbol{\Sigma}_{\text{n}} \mathbf{W}_{\text{new}}^T \mathbf{W}_{\text{new}}\right) \}$$

We then iterate over E and M steps until convergence

EM for PPCA: The Algorithm

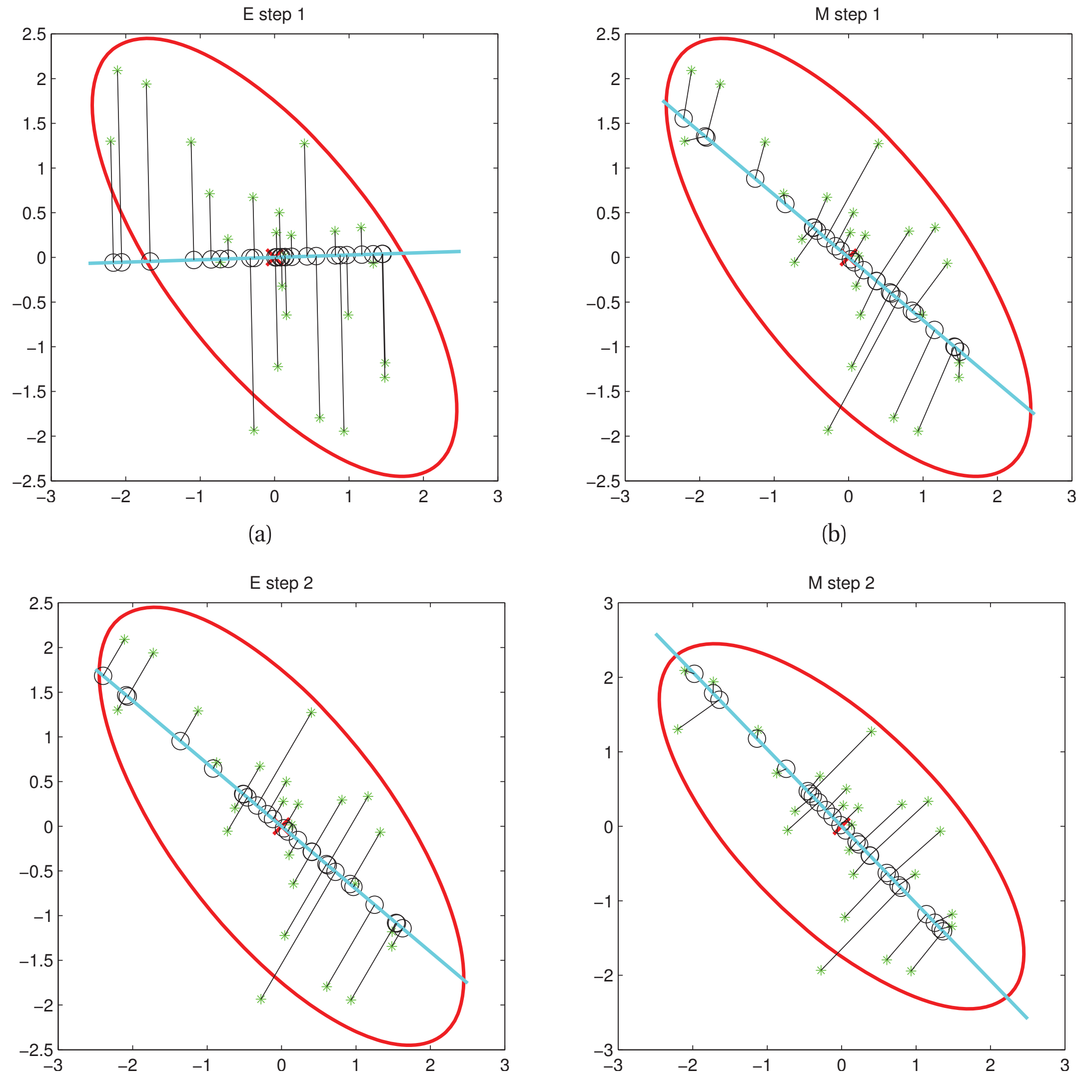
Goal: Given $\mathbf{X} = \{\mathbf{x}^{(n)}\}_{n=1}^{N}$ and *K* learn a PPCA model

Input: $\mathbf{X} = \{\mathbf{x}^{(n)}\}_{n=1}^{N}$ K

1. Initialise $\boldsymbol{\theta} = \{\mathbf{W}, \sigma^2\}$
2. Repeat until convergence
   a. E-step: Compute expected sufficient statistics $\mathbf{m}_n, \boldsymbol{\Sigma}_{\mathrm{n}}$ using current parameters $\boldsymbol{\theta} = \{\mathbf{W}, \sigma^2\}$
   b. M-step: Re-estimate $\boldsymbol{\theta} = \{\mathbf{W}, \sigma^2\}$ using current (fixed) expected sufficient statistics $\mathbf{m}_n, \boldsymbol{\Sigma}_{\mathrm{n}}$

EM for PPCA: Visualisation

Rigid rod and springs



(a) (b) (c) (d)

Advantages of EM for PPCA

- Computationally more efficient for large scale applications

Direct MLE

- Naïve MLE $\mathcal{O}(D^3)$
  - Top k eigenvectors: $\mathcal{O}(KD^2)$
- Evaluation of data covariance $\mathcal{O}(ND^2)$

EM

- Does not construct data covariance explicitly
- Most demanding step: $\mathcal{O}(NDK)$

- For large D and K << D this can be a significant saving and offset EM's iterative nature
- EM can be implemented in an online form
  - E-step can be computed for each data point separately
  - M-step can be done incrementally
  - Advantageous if both N and D are large

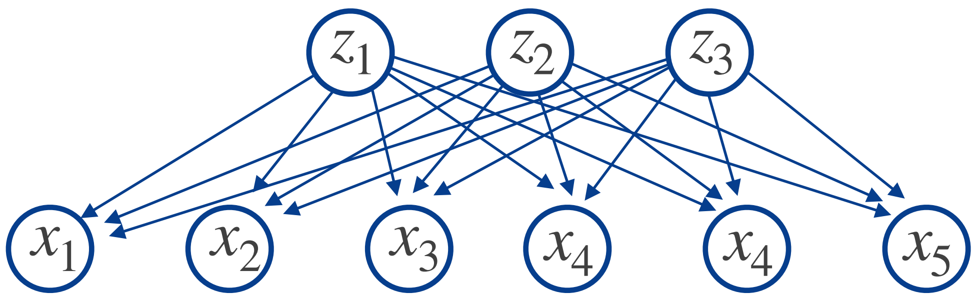
PCA & PPCA

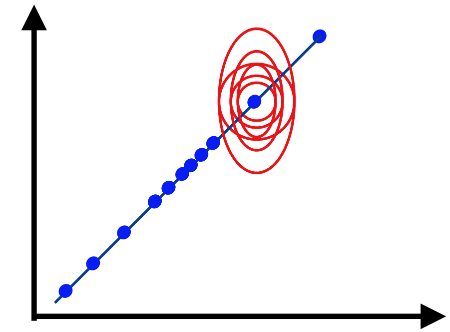
- PPCA is a constrained Gaussian model with covariance
  - $\mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}$
  - if K = D then $\mathbf{C} = \mathbf{S}$, the sample covariance
- The number of independent parameters only grows linearly with D (cf. full Gaussian, isotropic Gaussian)
  - But we still capture the K most significant correlations
- Rotational invariance → Statistical unidentifiability
- Assuming the data lies close to a hyperplane (linear technique)
  - Manifold may be curved or data can be clustered
- PCA is a very old technique with many different names, e.g. the Karhunen-Loève transform
  - Most commonly used dimensionality reduced method

III. Factor Analysis (FA)

Factor Analysis (FA)

$$\mathbf{X} = \{\mathbf{x}^{(n)}\}_{n=1}^{N} \qquad \mathbf{Z} = \{\mathbf{z}^{(n)}\}_{n=1}^{N}$$





PPCA	FA
$p(\mathbf{z}) = \mathcal{N}(\mathbf{z}\|\mathbf{0}, \mathbf{I}),$	$p(\mathbf{z}) = \mathcal{N}(\mathbf{z}\|\mathbf{0}, \mathbf{I}),$
$p(\mathbf{x}\|\mathbf{z}) = \mathcal{N}(\mathbf{x}\|\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I})$	$p(\mathbf{x}\|\mathbf{z}) = \mathcal{N}(\mathbf{x}\|\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \boldsymbol{\Psi})$

Diagonal instead of isotropic

Conditional independence (of features) still holds

Factor Analysis Conceptually

- As in PPCA, correlations induced by **W**
  - **W:** Factor loadings
  - Ψ: Uniquenesses (diagonal elements representing noise variances)

- Marginal likelihood is now: $p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{C})$ with:

$$\mathbf{C} = \mathbf{W}\mathbf{W}^T + \boldsymbol{\Psi}$$

- We can think of this as a low-rank approximation to the covariance matrix of **X:**
  - Parameters in full covariance Gaussian: $O(D^2)$
  - Parameters in **C:** $O(KD)$

Parameter Estimation in Factor Analysis (1)

- Maximum likelihood estimation
  - μ: Sample mean (as in PCA)
  - **W:** No closed-form solution → Need EM
- The parameters are: $\boldsymbol{\theta} = \{\boldsymbol{\mu}, \mathbf{W}, \boldsymbol{\Psi}\}$
- E-Step:

$$\mathbf{m}_n = \mathbb{E}_{p(\mathbf{z}^{(n)}|\mathbf{x}^{(n)},\boldsymbol{\theta}^{\text{old}})}\left[\mathbf{z}^{(n)}\right] = \mathbf{G}_{\text{old}}\mathbf{W}_{\text{old}}^T\boldsymbol{\Psi}_{\text{old}}^{-1}(\mathbf{x}^{(n)} - \bar{\mathbf{x}})$$

$$\boldsymbol{\Sigma}_{\text{n}} = \mathbb{E}_{p(\mathbf{z}^{(n)}|\mathbf{x}^{(n)},\boldsymbol{\theta}^{\text{old}})}\left[(\mathbf{z}^{(n)})(\mathbf{z}^{(n)})^T\right] = \mathbf{G}_{\text{old}} + \mathbf{m}_n\mathbf{m}_n^T$$

Where

$$\mathbf{G}_{\text{old}} = (\mathbf{I} + \mathbf{W}_{\text{old}}^T\boldsymbol{\Psi}_{\text{old}}^{-1}\mathbf{W}_{\text{old}})^{-1}$$

Note $\mathbf{G}_{\text{old}}$ is a *K x K* matrix

Parameter Estimation in Factor Analysis (2)

- M-step:

$$\mathbf{W}_{\text{new}} = \left[\sum_{n=1}^{N} (\mathbf{x}^{(n)} - \bar{\mathbf{x}})\mathbf{m}_n^T\right] \left[\sum_{n=1}^{N} \mathbf{\Sigma}_{\text{n}}\right]^{-1}$$

$$\mathbf{\Psi}_{\text{new}} = \text{diag}\{\mathbf{S} - \mathbf{W}_{\text{new}} \frac{1}{N} \sum_{n=1}^{N} \mathbf{m}_n (\mathbf{x}^{(n)} - \bar{\mathbf{x}})^T\}$$

- Where **S** is the sample covariance.

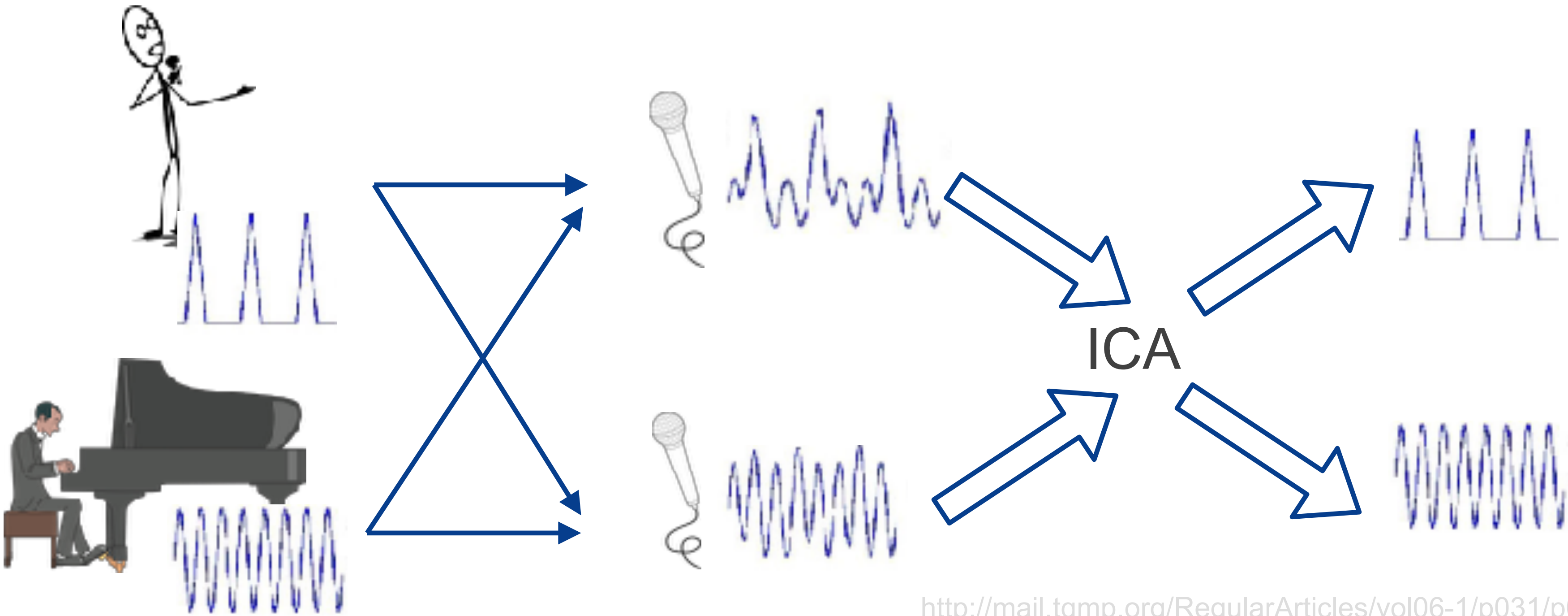
FA vs PPCA

PPCA	FA
Standard Normal (uncorrelated) prior p(**z**)	
Conditional likelihood p(**x** | **z**) is a linear mapping with isotropic noise	Conditional likelihood p(**x** | **z**) is a linear mapping with diagonal noise
Gaussian marginal p(**x**) with correlations determined by **W**	
Closed-form parameter estimation	EM required for parameter estimation
Solution invariant to rotations of the latent space	

IV. Independent Component Analysis (ICA)

The Cocktail Party Problem

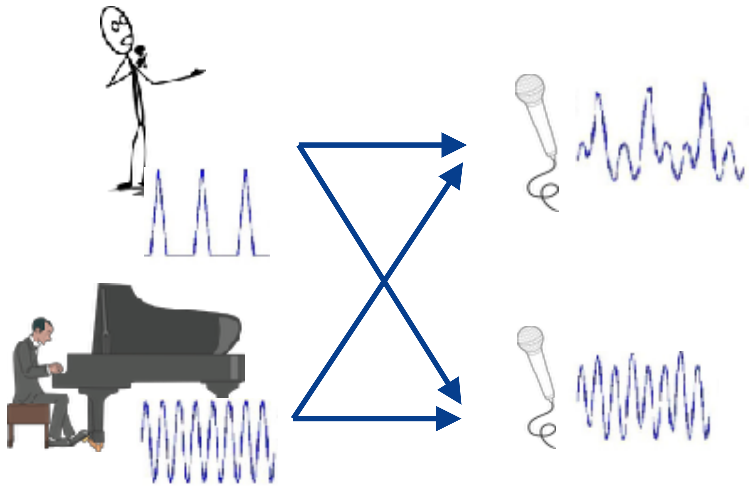
- You are in a crowded party and many people are speaking
- Your ears act as two microphones which are listening to a linear combination of the different speech signals
- Goal: Deconvolve the mixed signals into their constituents parts



http://mail.tqmp.org/RegularArticles/vol06-1/p031/p031.pdf

$$\mathbf{Z} = \{\mathbf{z}^{(n)}\}_{n=1}^{N}$$

$$\mathbf{z}^{(n)} \in \mathbb{R}^{K}$$



$$\mathbf{X} = \{\mathbf{x}^{(n)}\}_{n=1}^{N}$$

$$\mathbf{x}^{(n)} \in \mathbb{R}^{D}$$

$$\mathbf{z}^{(n)} \qquad \mathbf{W} \qquad = \mathbf{x}^{(n)} + \epsilon^{(n)}$$

$$\mathbf{x}^{(n)} = \boxed{\mathbf{W}}\mathbf{z}^{(n)} + \boxed{\boldsymbol{\epsilon}}^{(n)}$$

Mixing matrix

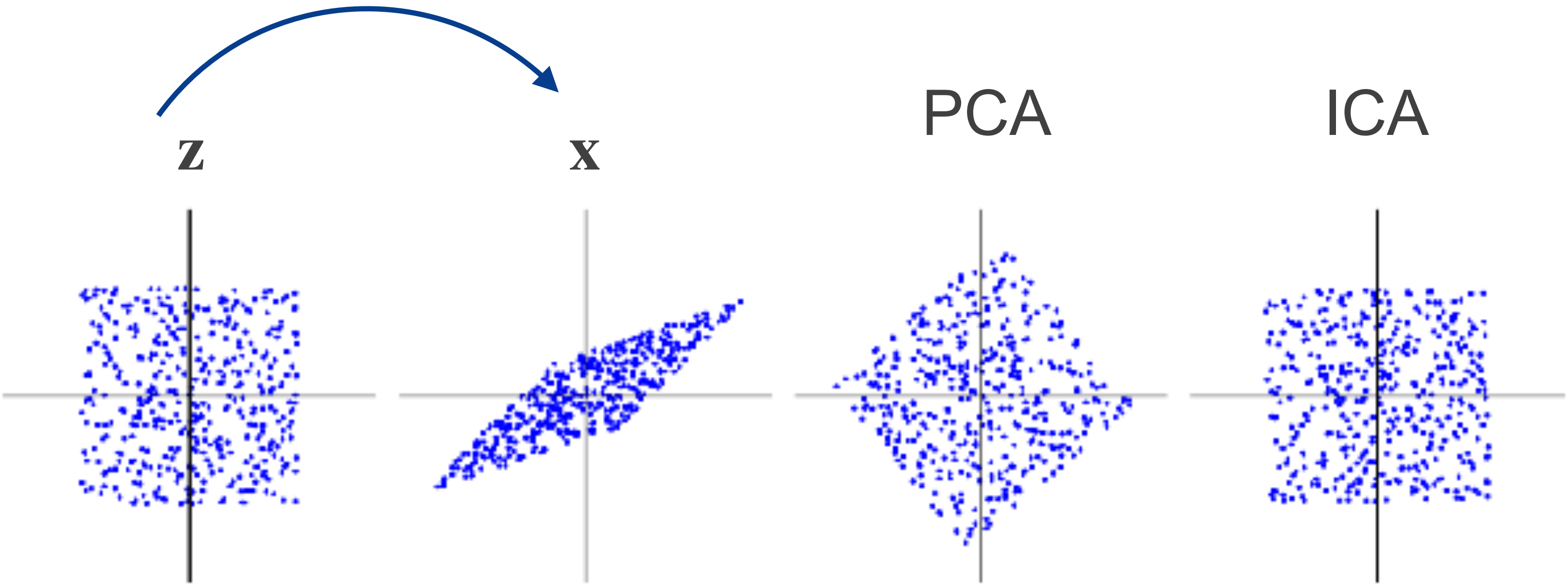
Noise model $\boldsymbol{\epsilon}^{(n)} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi})$

– Each data point treated independently (no direct temporal correlations)

Can PPCA to recover latent sources?

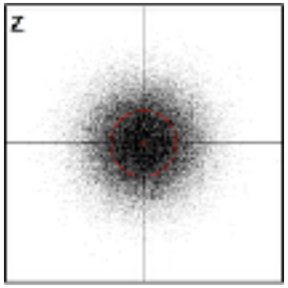
- No, because it is rotation invariant.

$$\mathbf{W} = \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}$$



Can PPCA to recover latent sources?

- No, because it is rotation invariant.



$$p(\mathbf{z}) = \mathcal{N}(\mathbf{z}|\mathbf{0}, \mathbf{I}),$$
$$p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}|\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I})$$

$$\Downarrow$$

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}),$$

If we multiply **W** by a rotation matrix, we still get same distribution over **x:**

$$\tilde{\mathbf{W}} = \mathbf{W}\mathbf{R} \qquad \tilde{\mathbf{W}}\tilde{\mathbf{W}}^T = \mathbf{W}\underbrace{\mathbf{R}\mathbf{R}^T}_{\mathbf{I}_{D\times D}}\mathbf{W}^T = \mathbf{W}\mathbf{W}^T$$

Can PPCA to recover latent sources?

$$\mathbf{x}^{(n)} = \boxed{\mathbf{W}}\mathbf{z}^{(n)} + \boxed{\boldsymbol{\epsilon}}^{(n)}$$

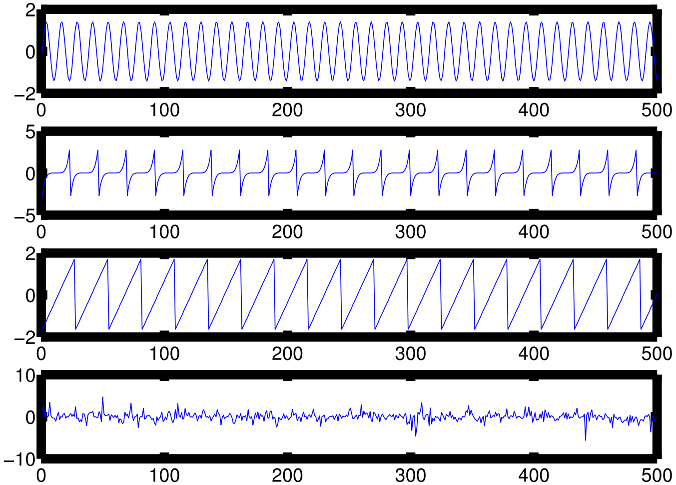
Mixing matrix

Noise model $\boldsymbol{\epsilon}^{(n)} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi})$

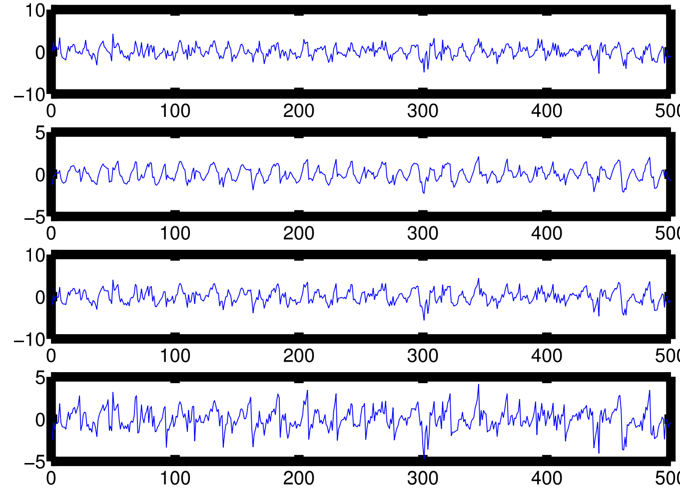
- PCA uses covariance matrix **S** which has D* (D+1)/2 elements but **W** has D * D elements
- We need distributions which have information beyond covariance matrices => non-Gaussian distributions

ICA Example (1)

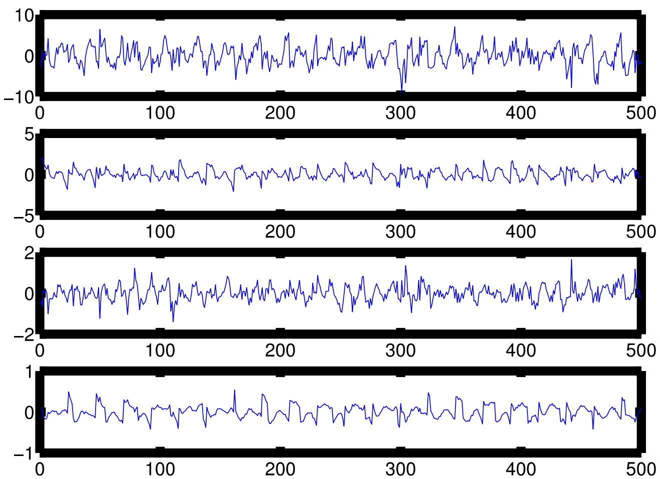
Truth



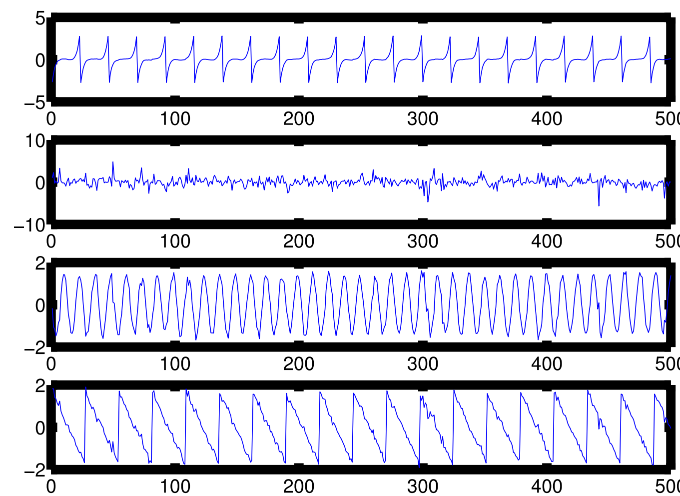
Observed signals



PCA Estimate

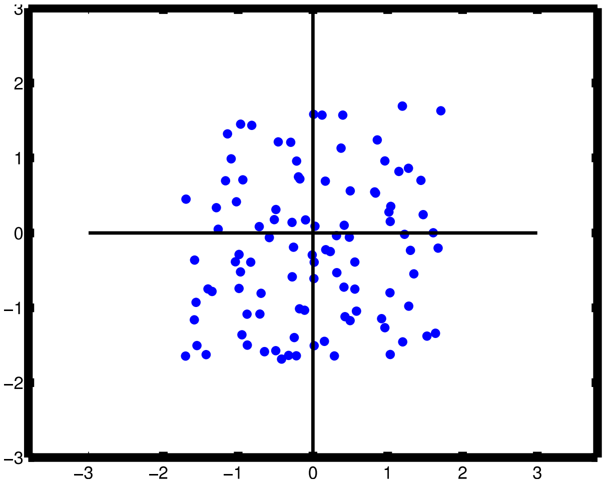


ICA Estimate

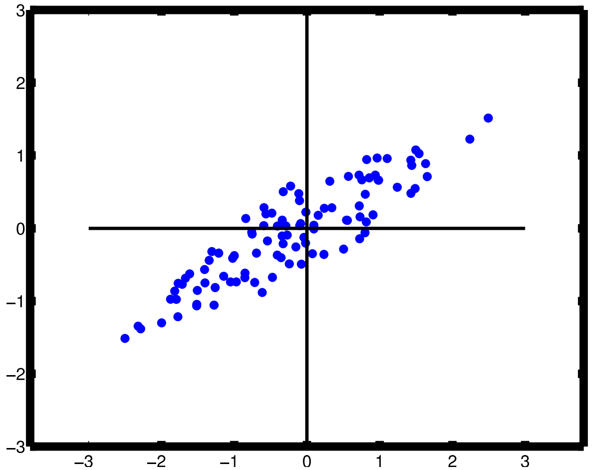


ICA Example 2

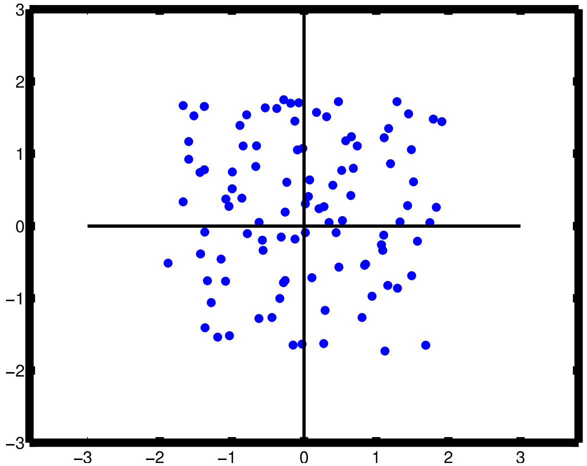
Uniform data



Mixed data



ICA Solution



- ICA can recover the source, up to a permutation of the indices and possible sign change
- Standard ICA requires **W** to be square and invertible
  - Otherwise (when having more sources than sensors), we cannot recover the true signal but can use posterior
- Need to estimate **W** (and possibly the source distribution)

http://wwwf.imperial.ac.uk/~nsjones/TalkSlides/HyvarinenSlides.pdf

ICA Model (2)

- Prior: Non-Gaussian independent prior

$$p(\mathbf{z}^{(n)}) = \prod_{k=1}^{K} \mathcal{N}(z_k^{(n)}; 0,1)$$

PPCA

$$p(\mathbf{z}^{(n)}) = \prod_{k=1}^{K} p_k(z_k^{(n)})$$

Source distribution

  - WLOG, can constrain variance of source distribution to be 1

- Usually assume noise level in the likelihood to be zero
  - Commonly known as noise-free ICA

- Goal: Infer p(**Z** | **X**)

- *Gaussian priors would not allow unique recovery of the sources*

MLE for ICA

Model: $$p(\mathbf{z}^{(n)}) = \prod_{k=1}^{K} p_k(z_k^{(n)})$$

$$\mathbf{x}^{(n)} = \mathbf{W}\mathbf{z}^{(n)} + \boldsymbol{\epsilon}^{(n)}$$

MLE: $$\arg\max_{\mathbf{W}} p(\mathscr{D} \mid \mathbf{W})$$

We need: $$P(\mathbf{x}^{(n)} \mid \mathbf{W})$$

Assuming: **W** is square and invertible
noise-free ICA

MLE for ICA

Noise-free ICA with square W

recognition weights

$$\mathbf{x} = \mathbf{W}\mathbf{z} \Rightarrow \mathbf{z} = \mathbf{W}^{-1}\mathbf{x} \Rightarrow \mathbf{z} = \mathbf{V}\mathbf{x}$$

- Since **x** = **W z** we have that:

$$\begin{aligned} p_x(\mathbf{x}) \\ &= p_x(\mathbf{W}\mathbf{z}) \\ &= p_z(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right| \\ &= p_z(\mathbf{z}) \left| \det \left(\mathbf{W}^{-1} \right) \right| \\ &= p_z(\mathbf{V}\mathbf{x}) \left| \det(\mathbf{V}) \right| \end{aligned}$$

MLE for ICA

Noise-free ICA with square W

$$p_x(\mathbf{W}\mathbf{z}) = p_z(\mathbf{z}) \left|\det(\mathbf{W}^{-1})\right| = p_z(\mathbf{V}\mathbf{x}) \left|\det(\mathbf{V})\right|$$

- Hence, assuming iid observations:

$$p(\mathcal{D}|\mathbf{V}) = \prod_{n=1}^{N} \prod_{k=1}^{K} p_k(\mathbf{v}_k^T \mathbf{x}^{(n)}) \left|\det(\mathbf{V})\right|$$

- Where $\mathbf{v}_k$ is the kth row of **V**

MLE for ICA

Noise-free ICA with square W

- Assume data have been centered and whitened (e.g. using PCA)

$$\mathbb{E}\left[\mathbf{x}\mathbf{x}^T\right] = \mathbf{I}$$

- Assuming zero-mean and unit-variance **z** (and noise-free observations):

$$\mathbb{C}\mathrm{ov}[\mathbf{x}] = \mathbb{E}\left[\mathbf{x}\mathbf{x}^T\right] = \mathbf{W}\mathbb{E}\left[\mathbf{z}\mathbf{z}^T\right]\mathbf{W}^T = \mathbf{W}\mathbf{W}^T$$

- Hence: **W W**$^{\mathrm{T}}$ = **I** → **W** is an orthogonal matrix

MLE for ICA (3)

Our goal is to minimize:

$$p(\mathcal{D}|\mathbf{V}) = \prod_{n=1}^{N} \prod_{k=1}^{K} p_k(\mathbf{v}_k^T \mathbf{x}^{(n)}) \, |\det(\mathbf{V})|$$

$$\begin{aligned}
\mathcal{L}(\mathbf{V}) &\stackrel{\text{def}}{=} -\frac{1}{N} \log p(\mathcal{D}|\mathbf{V}) \\
&= -\log|\det(\mathbf{V})| - \frac{1}{N} \sum_{k=1}^{K} \sum_{n=1}^{N} \log p_k(\mathbf{v}_k^T \mathbf{x}^{(n)}) \\
&= \sum_{k=1}^{K} \mathbb{E}[G_k(z_k)]
\end{aligned}$$

- Where $G_k(z) \stackrel{\text{def}}{=} -\log p_k(z)$ and $z_k = \mathbf{v}_k^T \mathbf{x}$
- Subject to the constraint **V** is an orthogonal matrix
- Gradient-based optimization
  - Standard first-order or second-order (Newton)
  - Alternatively, we can use EM

The Fast ICA Algorithm

- Approximate Newton method to optimise the average negative log likelihood $\mathcal{L}(\mathbf{V})$
- Uses an approximation to the Hessian

The Fast ICA Algorithm

- Approximate Newton method to optimise the average negative log likelihood $\mathcal{L}(\mathbf{V})$
- Uses an approximation to the Hessian

Assume only one latent factor z and that the source distribution is known, hence just write $G(\mathbf{z}) = -\log p(\mathbf{z})$

First and second derivatives of G(z)

- Updates:

$$\mathbf{v}^{\star} = \mathbb{E}\left[\mathbf{x}g'(\mathbf{v}^T\mathbf{x})\right] - \mathbb{E}\left[g''(\mathbf{v}^T\mathbf{x})\right]\mathbf{v}$$

$$\mathbf{v}^{\text{new}} = \frac{\mathbf{v}^{\star}}{\|\mathbf{v}^{\star}\|}$$

Projecting back onto the constraint surface

  - Expectations using Monte Carlo estimates from the training set
  - Iterate until convergence
  - Non-convex optimisation → multiple local minima

The Fast ICA Algorithm

- Approximate Newton method to optimise the average negative log likelihood $\mathcal{L}(\mathbf{V})$
- Uses an approximation to the Hessian

for p in 1…C

init $\mathbf{v}_p$ using a random vector

while $\mathbf{v}_p$ is not changing

$$\mathbf{v}_p = \mathbb{E}[\mathbf{x}g'(\mathbf{v_p}^T\mathbf{x})] - \mathbb{E}[g''(\mathbf{v_p}^T\mathbf{x})]\mathbf{v_p}$$

$$\mathbf{v_p} = \mathbf{v_p} - \sum_{j=1}^{p-1}\left[\mathbf{v_p}^T\mathbf{v_j}\mathbf{v_j}^T\right]$$

$$\mathbf{v}_p = \frac{\mathbf{v_p}}{||\mathbf{v_p}||}$$

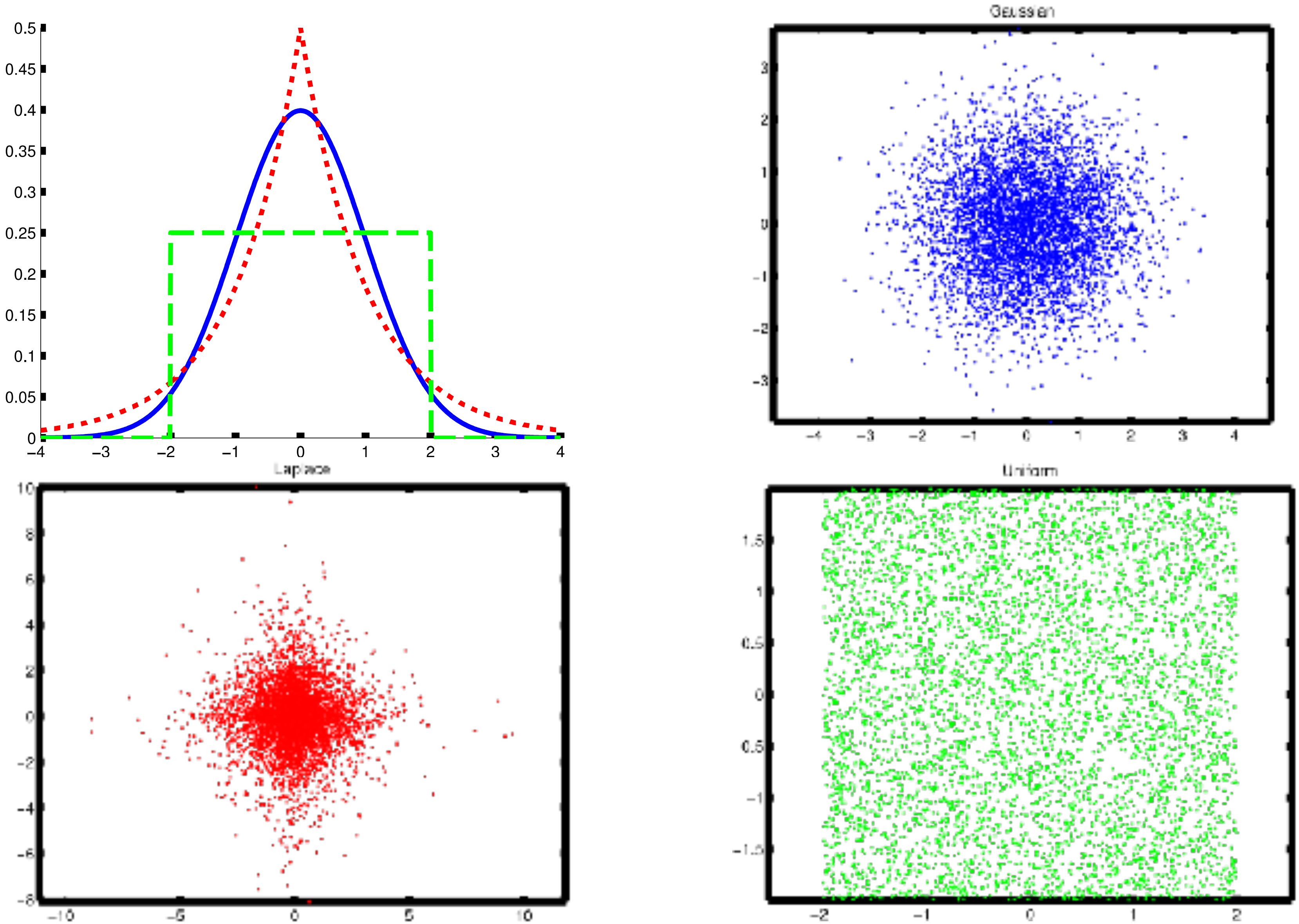
Source Distributions p(z)

- Gaussian is not an option
- Super-Gaussian distributions
  - More spiky peak and heavier tails, positive excess kurtosis
  - Laplace distribution

- Sub-Gaussian distributions
  - Negative excess kurtosis
  - Uniform distribution

- Skewed distributions
  - Measure of asymmetry
  - Gamma distribution

Examples of Source Distributions



Super-Gaussian Distributions for ICA

- Useful for natural signals such as images and speech
- Laplace prior seems obvious choice for source distribution
  - However, not differentiable at the origin: $\log p(z) = -\sqrt{2}|z| - \log(\sqrt{2})$
- Smoother options:
  - Logistic distribution
  - Hyperbolic secant distribution
- Shape of distribution need not to be known beforehand
  - Although it is important to know if it is super or sub Gaussian
- Can use simplified expression, for example:

$$G(z) = \log \cosh(z)$$

Other Estimation Principles for ICA

- Maximise non-Gaussianity
  - Maximise **negentropy**: Difference in (differential entropy) between Gaussian and source distribution

$$\text{negentropy}(z) \triangleq \mathbb{H}\left(\mathcal{N}(\mu, \sigma^2)\right) - \mathbb{H}\left(z\right)$$

- Minimise multi-information
  - KL divergence between joint distribution and factorised distribution

$$I(\mathbf{z}) \triangleq \mathbb{KL}\left(p(\mathbf{z})||\prod_j p(z_j)\right) = \sum_j \mathbb{H}(z_j) - \mathbb{H}(\mathbf{z})$$

- Infomax:
  - Maximise mutual information between **x** and **y**, where:

$$y_j = \phi(\mathbf{v}_j^T \mathbf{x}) + \epsilon$$

- All these criteria are closely related to maximum likelihood!

Conclusions

- Dimensionality reduction as a motivation for some continuous latent variable models
- PCA as the limit of a generative linear Gaussian model (PPCA) with a constrained covariance
  - Parameter estimation via direct likelihood maximisation or EM
- Factor analysis is very similar to PPCA but has a more general covariance structure
  - EM is required for parameter estimation
- ICA as a solution to cocktail-party problem by going beyond Gaussian prior distributions
- Reading:
  - Bishop (PRML, 2006): Ch 12 (except Sec 12.3, 12.4.2, 12.4.3)
  - Murphy (MLaPP, 2012): Ch 12 (except Sec 12.4, 12.5)